

Huanxiang Gao

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ACADEMIC DEGREES

Harbin Institute of Technology

- M.Sc. in Civil Engineering
 - GPA: 3.39/4.00 Rank: 15/119

Shenzhen, China
Sep. 2021 - Mar. 2024

Nanchang University

- B.Sc. in Civil Engineering
 - GPA: 3.74/4.00 Rank: 01/181

Nanchang, China
Sep. 2017 - Jun. 2021

EMPLOYMENT

National University of Singapore, Department of the Built Environment

- Research Associate (Building Robotics Lab)
 - Investigated a range of non-invasive physiological measurement devices.
 - Designing and implementing advanced computer vision and deep learning frameworks, we aim to accurately model the relationship between physiological measurements and thermal alliesthesia, thereby developing more intelligent and efficient indoor temperature control systems.

Singapore

Jun. 2024 - Present

Harbin Institute of Technology, Department of Civil & Environmental Engineering

- Research Assistant
 - Analyzed and determined the optimal configuration of wind field monitoring points affected by complex obstacles.
 - Developed methodologies for the rapid reconstruction of wind fields influenced by complex obstacles, utilizing sparse monitoring data.

China

Sep. 2021 - Mar. 2024

HONORS & AWARDS

- Harbin Institute of Technology **First-Class Scholarship**, 2021-2024.
- Nanchang University **First-Class Scholarship**, 2017-2021.
- **Third Prize** in the 12th Chinese Mechanics Competition, 2019.
- **Excellence prize** in the 12th Chinese Basic Mechanics Experiment (team competition), 2019.
- **First Prize** in Jiangxi Division of National Mathematical Contest in Modeling, 2019.

RESEARCH INTERESTS

- Personal indoor thermal comfort
- Thermal alliesthesia
- Deep learning
- Non-intrusive sensing
- Computational Fluid Dynamics
- Computer vision

SKILLS

- Programming Languages & Environments: Proficient in Python and MATLAB; experienced with Linux operating system.
- Data Science & Libraries: Expertise in Python libraries including Scikit-Learn, NumPy, SciPy, Matplotlib, and PyTorch for advanced data analysis and deep learning applications. Specialized in designing physics-informed and time-series neural networks.
- Computational Fluid Dynamics: Experienced with OpenFOAM for fluid dynamic simulations.
- Geometric Modeling: Skilled in using AutoCAD and Sharp3D.

EXTRACURRICULAR ACTIVITIES

Member of the Organization Department of the Nanchang University Student Union

- Responsible for organizing school activities, procuring materials, and coordinating personnel schedules.

Member of the Harbin Institute of Technology Wild Cat Rescue Association

- Responsible for fundraising for wild cat rescue, accompanying cats to the vet, and has rescued 4 cats.

Community volunteer

- Assisted with nucleic acid testing and distributed emergency food supplies during the COVID-19 period.

PUBLICATIONS

1. **H.X. Gao**, G. Hu*, D.Q. Zhang, W.J. Jiang, K.T. Tse, Bernd R. Noack, 2024. SiGAN: A 3D sensor importance deep generative model for urban wind flow field monitoring. *Building and Environment* (JCR Q1, IF=7.4).
2. **H.X. Gao**, G. Hu*, D.Q. Zhang, W.J. Jiang, K.T. Tse, K.C.S. Kwok*, A. Kareem, 2024. Urban Wind Field Prediction Based on Sparse Sensors and Physics-informed Graph-assisted Auto-Encoder. *Computer-Aided Civil and Infrastructure Engineering* (JCR Q1, IF=9.6).
3. **H.X. Gao**, G. Hu*, D.Q. Zhang, W.J. Jiang, W.L. Chen*, 2024. Multi-elevation Mountain Wind Field Prediction Based on Deep Learning. *Applied Energy* (JCR Q1, IF=11.2).
4. Y.R. Li, Y. Lei, **H.X. Gao**, Hu, G.*, 2024. A machine learning-augmented aerodynamic database of rectangular cylinders. *Physics of Fluids*.
5. W.J. Jiang, B. Liu, Y. Liang, **H.X. Gao**, P.F. Lin, D.Q. Zhang, G. Hu, 2024. Applicability analysis of transformer to wind speed forecasting by a novel deep learning framework with multiple atmospheric variables *Applied Energy*.
6. D.Q. Zhang, B. Liu, Y. Liang, W.J. Jiang, **H.X. Gao**, J.Z. Zhang, G. Hu, 2024. Numerical study of dynamic amplification factor and characteristic wind curves of high-speed train in tornado-like vortices *Journal of Wind Engineering and Industrial Aerodynamics*.
7. W.J. Jiang, J.Z. Zhang, Y.R. Li, D.Q. Zhang, G. Hu, **H.X. Gao**, Z.D. Duan, 2024. Advancing storm surge forecasting from scarce observation data: A causal-inference based Spatio-Temporal Graph Neural Network approach *Coastal Engineering*.
8. **H.X. Gao**, J.L. Liu, P.F. Lin, G.* Hu, L. Patruno, Y.Q. Xiao, K.T. Tse, K.C.S. Kwok, 2023. An Optimal Sensor Placement Scheme for Wind Flow and Pressure Field Monitoring. *Building and Environment* (JCR Q1, IF=7.4).
9. D.Q. Zhang, Z.Q. Liu, X.Y. Jiang, W.J. Jiang, **H.X. Gao**, Hu, G.*, , C. Li, Y.Q. Xiao, G. Hu*, 2023. Numerical study of flow characteristics of tornado-like vortices considering both swirl ratio and aspect ratio. *Journal of Wind Engineering and Industrial Aerodynamics*.
10. D.Q. Zhang, G. Hu, J. Song, **H.X. Gao**, H. Ren, W.L. Chen, 2023. A novel spatio-temporal wind speed forecasting method based on the microscale meteorological model and a hybrid deep learning model *Energy*.
11. W.J. Jiang, P.F. Lin, Y. Liang, **H.X. Gao**, D.Q. Zhang, G. Hu, 2023. A novel hybrid deep learning model for multi-step wind speed forecasting considering pairwise dependencies among multiple atmospheric variables *Energy*.
12. **H.X. Gao**, J.L.* Liu, P.F. Lin**, C. Li, Y.Q. Xiao, G. Hu, 2022. Pedestrian Level Wind Flow Field of Elevated Tall Buildings with Dense Tandem Arrangement. *Building and Environment* (JCR Q1, IF=7.4).